

Project Initialization and Planning Phase

Date	24 June 2026
Team ID	SWTD-2026-3249
Project Title	OptiCrop - Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report outlines how OptiCrop applies machine learning to transform agricultural production planning, boosting yield, resource efficiency, and farmer income. It tackles the lack of timely, localized farming guidance, promising smarter irrigation and fertilization, reduced input waste, and improved harvest outcomes. Key features include a machine learning-based crop recommendation model and real-time, weather-aware advisory generation.

Project Overview	
Objective	The primary objective is to optimize agricultural production by using machine learning to generate real-time, field-specific recommendations for irrigation, fertilization, and crop planning, improving yield and resource efficiency.
Scope	The project covers soil and weather data ingestion, crop-yield prediction, irrigation and fertilizer optimization, and a farmer-facing mobile/web advisory system, adaptable across multiple crops, regions, and growing seasons.
Problem Statement	
Description	Smallholder farmers often lack timely, localized guidance on irrigation, fertilization, and crop selection, leading to overuse or underuse of resources, inconsistent yields, and avoidable financial losses.
Impact	Solving this will increase crop yields and farmer income, reduce water and fertilizer overuse and its environmental impact, and improve food security and rural livelihoods.
Proposed Solution	
Approach	Employing machine learning models trained on soil composition, historical weather patterns, and crop-specific requirements to generate dynamic, localized recommendations for irrigation, fertilization, pest control, and crop selection.
Key Features	- Implementation of a machine learning-based crop yield prediction and recommendation model.
	- Real-time, weather-aware decision-making for timely irrigation and fertilization. - Continuous learning from farmer feedback and seasonal outcomes to refine recommendations.

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	T4 GPU
Memory	RAM specifications	16 GB
Storage	Disk space for data, models, and logs	1 TB SSD
Software		
Frameworks	Python frameworks	Django REST Framework, Flask
Libraries	Additional libraries	scikit-learn, pandas, numpy, TensorFlow, matplotlib, seaborn, GeoPandas
Development Environment	IDE	Jupyter Notebook, VS Code
Data		
Data	Source, size, format	Kaggle crop-recommendation dataset, 2200 records, csv; IMD weather data, csv; Soil Health Card dataset, csv